

Name _____ Period _____

Skill 29.1 Exercise 1	
label each pair of events as independent or dependent and give a one-sentence reason	
A coin is flipped twice. Event A: first flip is heads. Event B: second flip is heads.	
Coin flipped twice (first flip heads, second flip heads) Answer: Independent	
A card is drawn from a deck, <u>not</u> replaced, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart	
Card drawn, not replaced, then second card drawn (first heart, second heart) Answer: Dependent Reason: Not replacing the card changes the deck composition.	
A card is drawn, replaced, shuffled, then a second card is drawn. Event A: first card is a heart. Event B: second card is a heart.	
Card drawn, replaced, shuffled, then second card drawn (first heart, second heart) Answer: Independent Reason: Replacing and shuffling resets the deck to the original conditions.	
A spinner with equal sections is spun twice. Event A: first spin lands on blue. Event B: second spin lands on blue.	
Spinner spun twice (first blue, second blue) Answer: Independent Reason: One spin does not affect the next spin.	
A bag has 4 green and 6 yellow marbles. Two marbles are drawn without replacement. Event A: first marble is green. Event B: second marble is green.	
Bag with 4 green and 6 yellow marbles, two draws without replacement (first green, second green) Answer: Dependent Reason: Without replacement, the first draw changes what remains in the bag.	

Skill 29.2 Exercise 1	
label each pair of events as mutually exclusive or not mutually exclusive and give a short reason.	
Event A: roll an even number or Event B: roll an odd number	Answer: Mutually exclusive Reason: A number cannot be both even and odd on one roll. Intersection: $\emptyset$
Event A: roll a number greater than 4 or Event B: roll an even number	A={5,6} B={2,4,6} Answer: Not mutually exclusive Reason: They overlap at 6. Intersection: {6}
Event A: roll a 2 or Event B: roll a number less than 4	A={2} B={1,2,3} Answer: Not mutually exclusive Reason: If you roll 2, both events happen. Intersection: {2}
Event A: roll a prime number or Event B: roll an odd number	Primes on a die: {2,3,5} Odd: {1,3,5} Answer: Not mutually exclusive Reason: They overlap at 3 and 5. Intersection: {3,5}

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Skill 29.3 Exercise 1	
Calculate the probability of rolling a prime number or rolling an odd number by completing the steps.	
Step 1: Is this a mutually exclusive or not mutually exclusive?	The situation is <u>not mutually exclusive</u> because you can roll a prime and an odd number $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$
Step 2: What is $P(A)$?	All the possible prime numbers are {1, 2, 3, 5} $P(A) = 4/6$
Step 3: What is $P(B)$?	All the possible odd numbers are {1, 3, 5} $P(B) = 3/6$
Step 4: What is $P(A \text{ and } B)$?	$P(A \text{ and } B)$ is the intersection of $P(A)$ and $P(B)$. Intersection values = {1, 3, 5} $P(A \text{ and } B) = 3/6$
Step 5: What is $P(A \text{ or } B)$?	$P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B) = 4/6 + 3/6 - 3/6 = 4/6$ Conclusion: There is a 67% probability of rolling a prime number OR a number that is odd

Skill 29.4 Exercise 1
A bag contains 3 red marbles and 2 blue marbles. Two marbles are drawn without replacement . Write a conditional probability expression (using notation) for each statement: a. The probability that the second marble is red , given that the first marble is blue b. The probability that the second marble is blue , given that the first marble is red $P(\text{Red Second} \mid \text{Blue First})$ $P(\text{Blue Second} \mid \text{Red First})$
For each statement above, calculate the probability. $P(\text{Red Second} \mid \text{Blue First}) = 3/4$ $P(\text{Blue Second} \mid \text{Red First}) = 2/4$

Skill 29.4 Exercise 2
A bag contains 3 red marbles and 2 blue marbles. Two marbles are drawn with replacement . Write a conditional probability expression (using notation) for each statement: a. The probability that the second marble is red , given that the first marble is blue b. The probability that the second marble is blue , given that the first marble is red $P(\text{Red Second} \mid \text{Blue First}) = P(\text{Red})$ $P(\text{Blue Second} \mid \text{Red First}) = P(\text{Blue})$
For each statement above, calculate the probability. $P(\text{Red}) = 3/5$ $P(\text{Blue}) = 2/5$

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Skill 29.5 Exercise 1

A bag has 3 red and 2 blue marbles. Two marbles are drawn **without replacement**. Write an expression for: *The probability the first is red **and** the second is blue*

$$P(\text{Red 1}^{\text{st}} \text{ and Blue 2}^{\text{nd}}) = P(\text{Red 1st}) \cdot P(\text{Blue 2}^{\text{nd}} | \text{Red 1}^{\text{st}})$$

Compute marble probabilities (without replacement). Using the same bag (3 red, 2 blue), two draws without replacement.

Calculate:

- $P(\text{Blue 1st and Blue 2nd})$
- $P(\text{Blue 1st and Red 2nd})$

$$BB = 2/5 \cdot 1/4 = 1/10$$

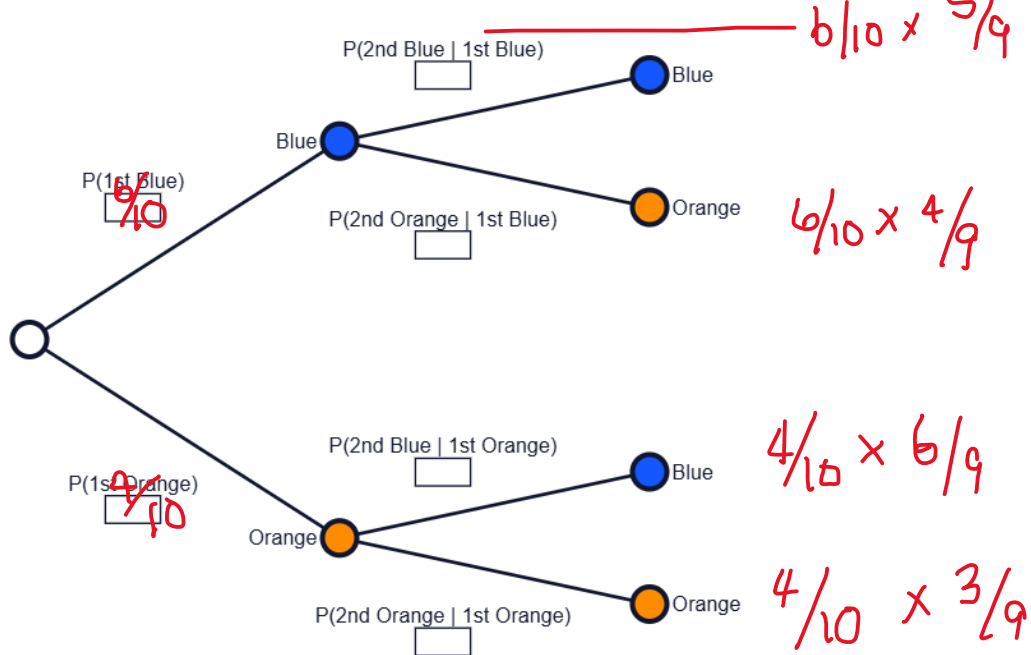
$$BR = 2/5 \cdot 3/4 = 3/10$$

Compute the probabilities with replacement using the same bag.

$$P(A \text{ and } B) = P(A) \cdot P(A | B) = P(A) \cdot P(B)$$

$$P(A \text{ and } B) = 2/5 \cdot 2/5 = 4/25$$

$$P(A \text{ and } B) = 2/5 \cdot 3/5 = 6/25$$

[illegible]

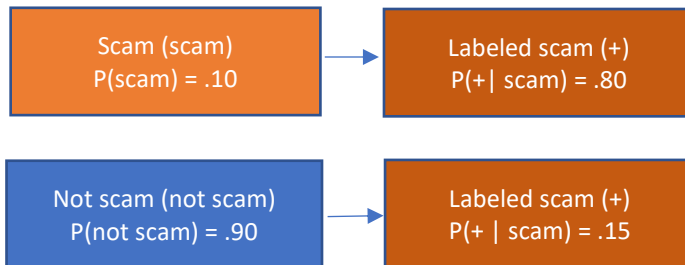
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Skill 29.7 Exercise 1

Your phone uses a filter to label text messages as “Scam” or “Not Scam.” The filter is helpful, but it is not perfect. Suppose the following are true,

- 10% of all messages are actually scams ($P(\text{scam}) = .10$)
- 90% of all messages are not scams ($P(\text{not scam}) = .90$)
- If a message is a scam, the filter labels it a scam 80% of the time ($P(\text{label} | \text{scam}) = .80$)
- If a message is not a scam, the filter still labels it as a scam 15% of the time ($P(\text{label} | \text{not scam}) = .15$)

These outcomes are shown visually below.



If your phone labels a message as “Scam,” what is the probability that the message is actually a scam? This is, $P(\text{scam} | +)$

Consider 100 messages. Based on the probabilities above, how many are scams? How many are not scams?

- 10 are scams
- 90 are not scams

Based on the probabilities above, how many of the *scams* are correctly labeled *scam*?

80% of 10 is 8

Based on the probabilities above, how many of the *not scams* are incorrectly labeled as *scam*?

15% of 90 is 13.5

What is the total messages that are labeled as *scam*?

8 actual scams + 13.5 non scams

What is $P(\text{scam} | +)$. That is, out of the messages labeled as scam (+), how many were actually scams?

$8/21 = .37$